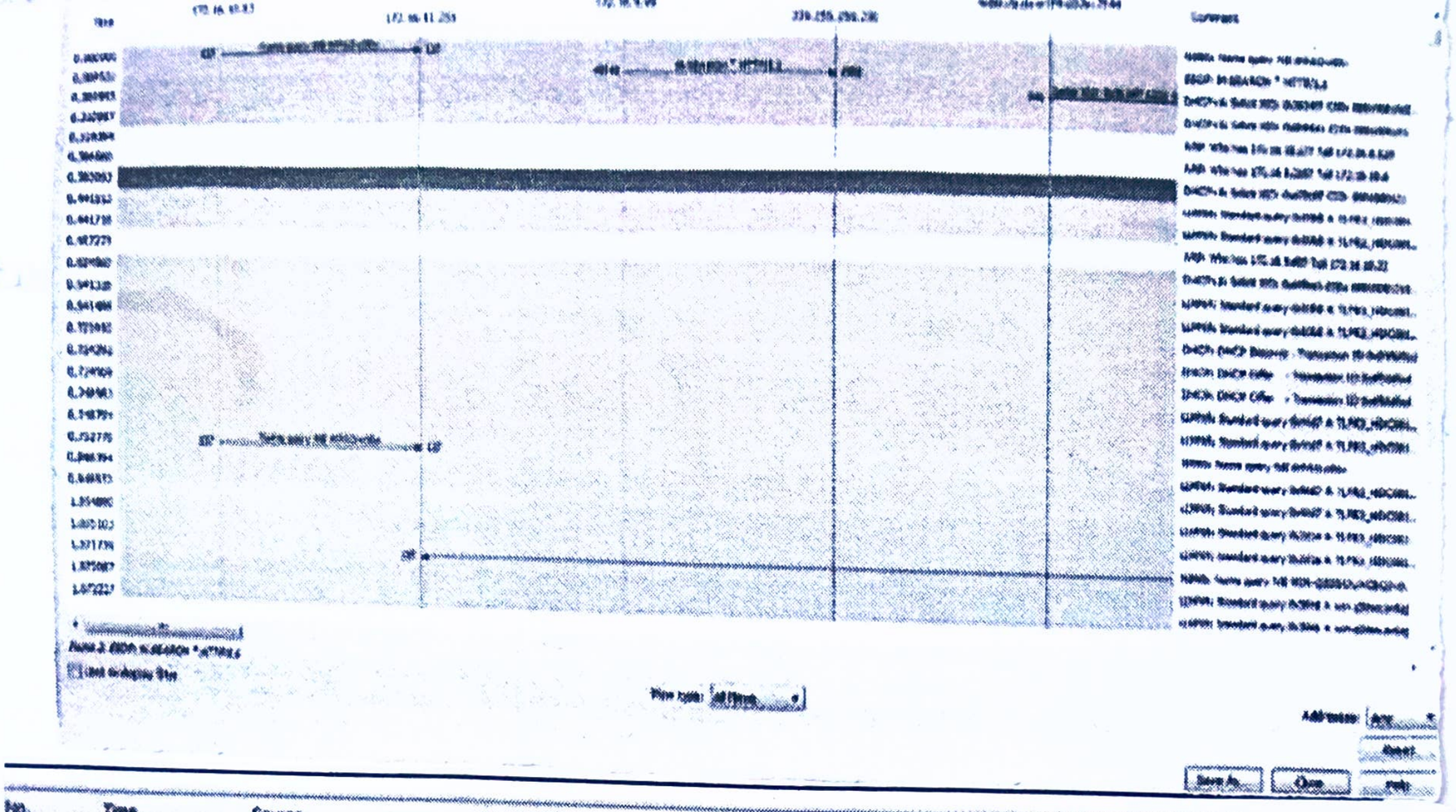


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Pegatron_e0:87:9e	Broadcast	ARP	60	Who has 172.16.9.94? Tell 172.16.9.138
2	0.000180	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.10.36? Tell 172.16.10.50
3	0.000294	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.36? Tell 172.16.10.50
4	0.000295	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.8.37? Tell 172.16.10.50
5	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.9.37? Tell 172.16.10.50
6	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.37? Tell 172.16.10.50
7	0.001460	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0xae2b A TLFL3-HDC101701
8	0.001622	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0xae2b A TLFL3-HDC101701
9	0.001623	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0x28c8 AAAA TLFL3-HDC101701
10	0.001625	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0x28c8 AAAA TLFL3-HDC101701

```
▸ Frame 7: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
▸ Ethernet II, Src: Dell_35:10:a8 (50:9a:4c:35:10:a8), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
▾ User Datagram Protocol, Src Port: 62374, Dst Port: 5355
    Source Port: 62374
    Destination Port: 5355
    Length: 41
    Checksum: 0x90e0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
▸ Link-local Multicast Name Resolution (query)
```

```
0000  33 33 00 01 00 03 50 9a  4c 35 10 a8 86 dd 60 00   33....P. L5....`.
0010  00 00 00 29 11 01 fe 80  00 00 00 00 00 00 49 68   ...).... ......Ih
0020  12 a7 5e 36 52 3e ff 02  00 00 00 00 00 00 00 00   ..^6R>.. ........
0030  00 00 00 01 00 03 f3 a6  14 eb 00 29 90 e0 ae 2b   ........ ...)...+
0040  00 00 00 01 00 00 00 00  00 00                     ........ ..
```

Wireshark · Follow TCP Stream (tcp.stream eq 35) · wireshark_D32D7F…

```
..........a5b@M.......b....`Z.y'.........
.8..|..../..........V.......`....jj.+./...0.............../.5.
...|jj.................login.live.com.....#...
*
..:i.....}}..........U..Y:..UL5.].U:...85(.............K. .9..P...&.}
b....F..$|...|.....E.....
+.............
..
...r0..n0..V.......~.....ltm.x.K..H0
*         *.H..
.....0w1.0  ..U....US1.0...U.
..Symantec Corporation1.0...U....Symantec Trust
Network1(0&..U....Symantec Class 3 EV SSL CA - G30..
151120000000Z.
171202359590Z0...1.0...4.....7<.....US1.0...+.....7<....
Washington1.0...U....Private Organization1.0...U...      6004134851.0
```

7 client pkts, 9 server pkts, 7 turns.

Entire conversation (12 kB) | Show and save data as ASCII | Stream 35

Find: | Find Next

Filter Out This Stream | Print | Save as... | Back | Close | Help

Packets: 2137 · Displayed: 21 (1.0%)

EX:-05

E:-

7|26

EXPERIMENT ON PACKET CAPTURE TOOL

WIRE SHARK

AIM:-

To understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

PACKET SNIFFER :-

* Capture packets sent and received by the computer.
* Displays protocol information of captured packets.
* Works passively.

PACKET SWIFFER TOOLS :-

* TCP dump - commaand line packet capture tool.
* wire shark - Graphical packet analyser.

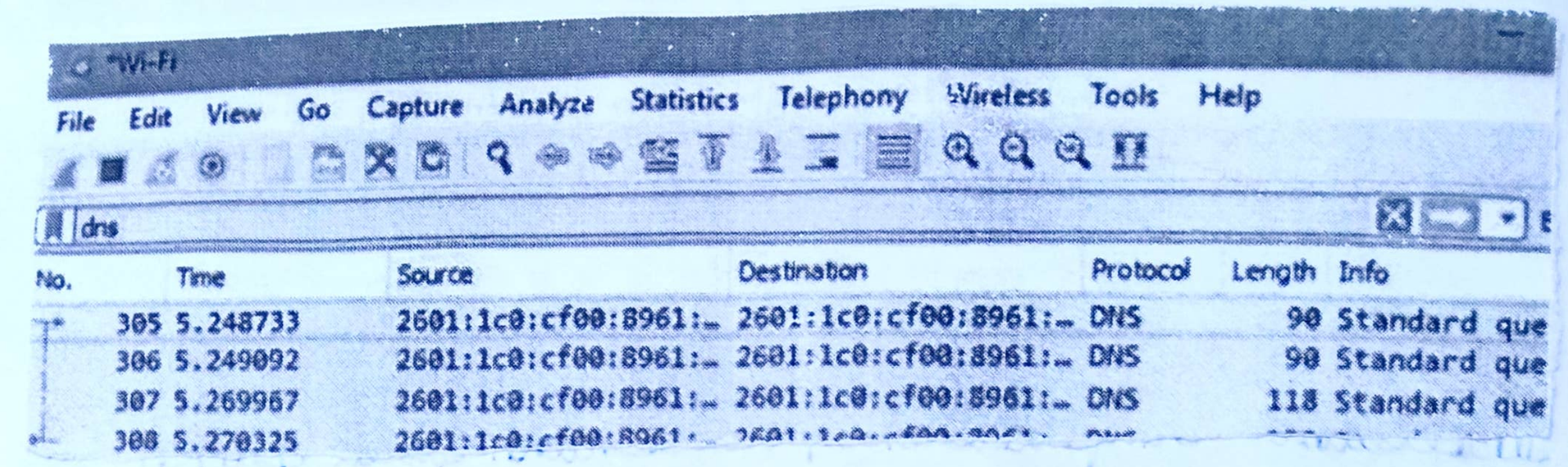
DESCRIPTION :-

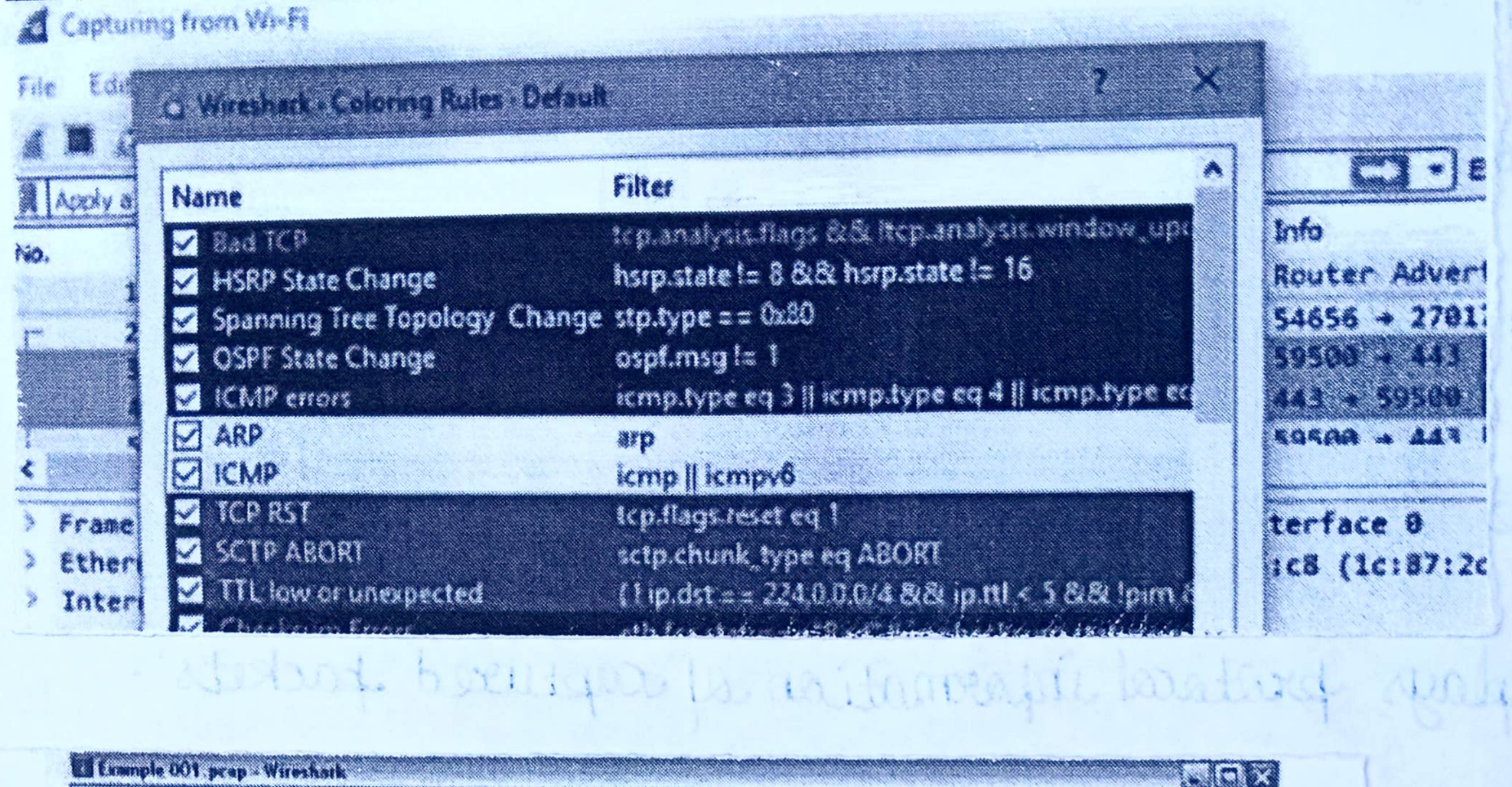
Wireshark is a network protocol analyser that captures and displays networks packets in real time.

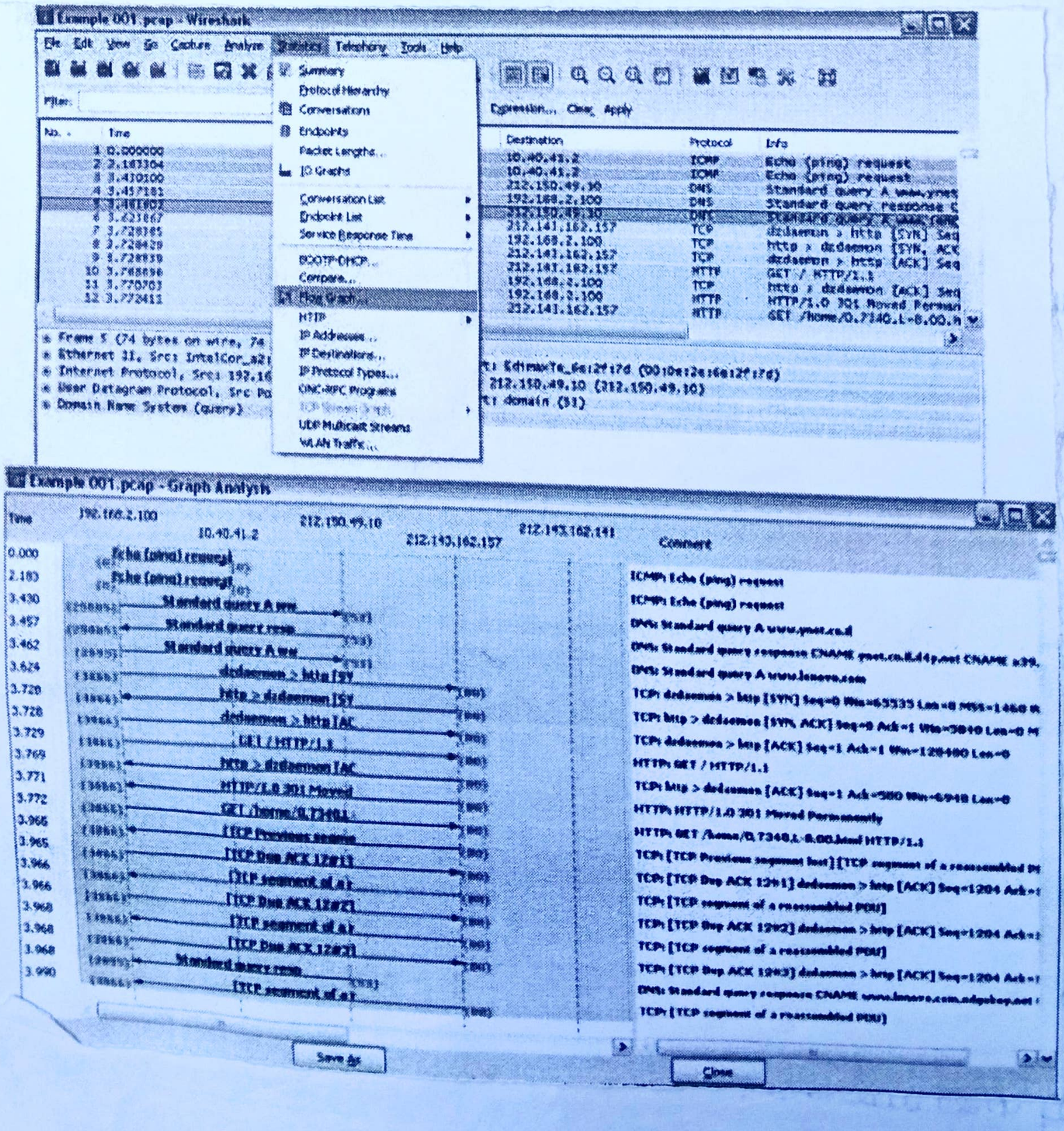
It ~~http~~ helps monitor, analyse, troubleshoot and inspect network traffic.

Uses of Networks :-

* Captures network traffic.
* Analyses protocols.
* Applies display filters.


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
305 5.248733 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 90 Standard que
306 5.249092 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 90 Standard que
307 5.269967 2601:1c0:cf00:8961:… 2601:1c0:cf00:8961:… DNS 118 Standard que
308 5.270325


Capturing from Wi-Fi
Wireshark · Coloring Rules · Default
Name Filter
Bad TCP tcp.analysis.flags && !tcp.analysis.window_upd
HSRP State Change hsrp.state != 8 && hsrp.state != 16
Spanning Tree Topology Change stp.type == 0x80
OSPF State Change ospf.msg != 1
ICMP errors icmp.type eq 3 || icmp.type eq 4 || icmp.type eq
ARP arp
ICMP icmp || icmpv6
TCP RST tcp.flags.reset eq 1
SCTP ABORT sctp.chunk_type eq ABORT
TTL low or unexpected (!ip.dst == 224.0.0.0/4 && ip.ttl < 5 && !pim
Info
Router Advert
54656 → 2781
59500 → 443
443 → 59500
terface 0
:c8 (1c:87:2c


Example 001.pcap - Wireshark
File Edit View Go Capture Analyze Statistics Telephony Tools Help
Summary
Protocol Hierarchy
Conversations
Endpoints
Packet Lengths...
IO Graphs
Conversation List
Endpoint List
Service Response Time
BOOTP-DHCP...
Compare...
Flow Graph...
HTTP
IP Addresses...
IP Destinations...
IP Protocol Types...
ONC-RPC Programs
UDP Multicast Streams
WLAN Traffic...
Destination Protocol Info
Example 001.pcap - Graph Analysis
Time 192.168.2.100 10.40.41.2 212.150.49.10 212.141.162.157 212.193.162.141 Comment
ICMP: Echo (ping) request
ICMP: Echo (ping) request
DNS: Standard query A www.ynet.co.il
DNS: Standard query response CNAME ynet.co.il.d4p.net CNAME a39.
DNS: Standard query A www.lenovo.com
TCP: dnsconnect > http [SYN] Seq=0 Win=65535 Len=0 MSS=1460 W
TCP: http > dnsconnect [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 M
TCP: dnsconnect > http [ACK] Seq=1 Ack=1 Win=128480 Len=0
HTTP: GET / HTTP/1.1
TCP: http > dnsconnect [ACK] Seq=1 Ack=380 Win=6948 Len=0
HTTP: HTTP/1.0 301 Moved Permanently
HTTP: GET /home/0,7340,L-8,00.html HTTP/1.1
TCP: [TCP Previous segment lost] [TCP segment of a reassembled PD
TCP: [TCP Dup ACK 1241] dnsconnect > http [ACK] Seq=1204 Ack=
TCP: [TCP segment of a reassembled PDU]
TCP: [TCP Dup ACK 1242] dnsconnect > http [ACK] Seq=1204 Ack=
TCP: [TCP segment of a reassembled PDU]
TCP: [TCP Dup ACK 1243] dnsconnect > http [ACK] Seq=1204 Ack=
DNS: Standard query response CNAME www.lenovo.com.edgekey.net
TCP: [TCP segment of a reassembled PDU]
Save As
Close

Applications :-

* Network troubleshooting.
* Security analysis.
* Protocol debugging.
* Learning network protocols.

Capturing Packet :-

1. Open wireshark.
2. Select the network interface.
3. Click start capture.
4. Observe packet in real time.
5. Click stop and Save the capture.

Wireshark packet :-

* Packet list.
* Packet details.
* Packet bytes

Color coding :-

* purple : TCP
* Blue : UDP
* Black : Error packets.

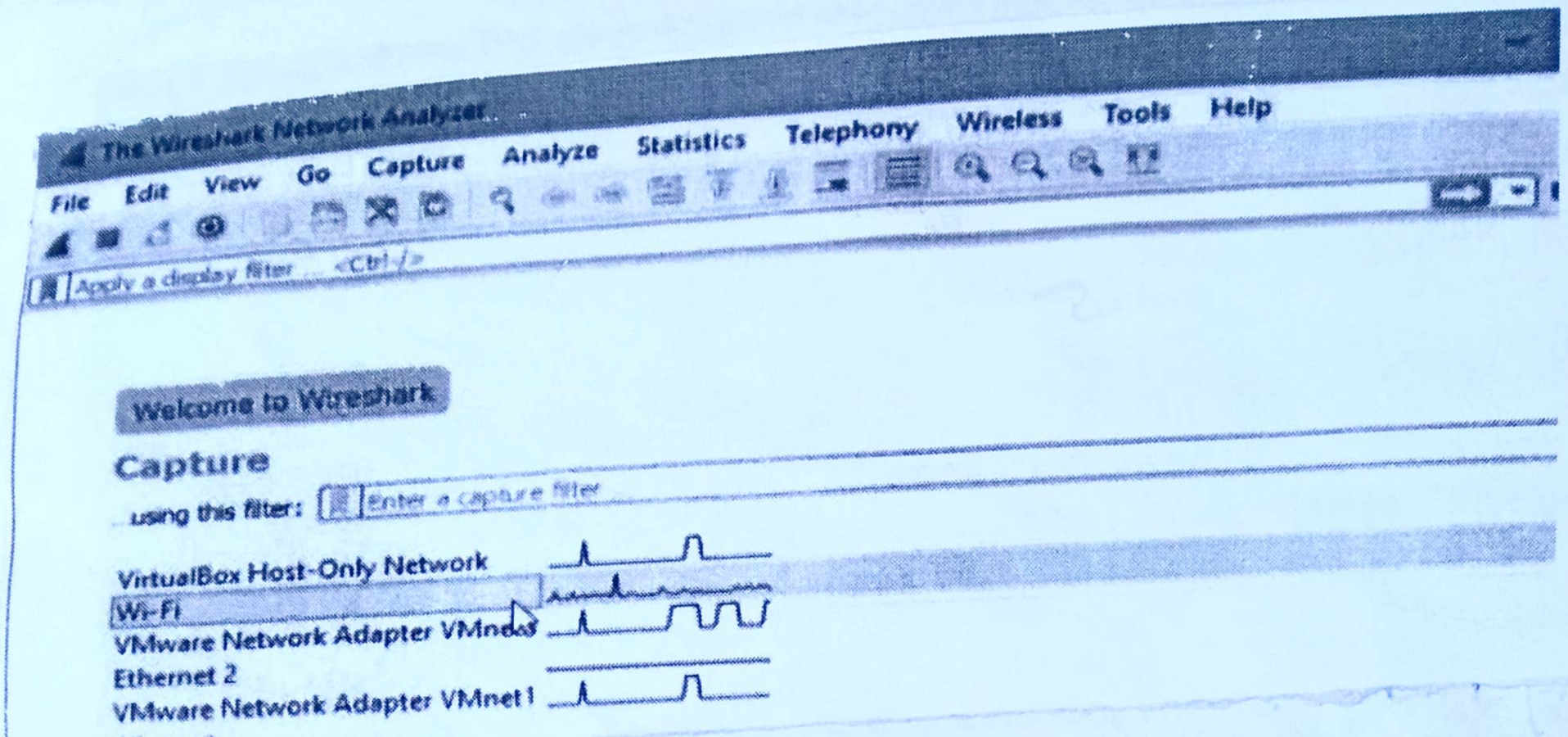
Filtering Packets :-

* Enter a filter in the filter bar.
* Right click to apply filter

Flow Graph :-

Displays communication flow between devices, making packing exchange easier to understand

Capture 100 packets.



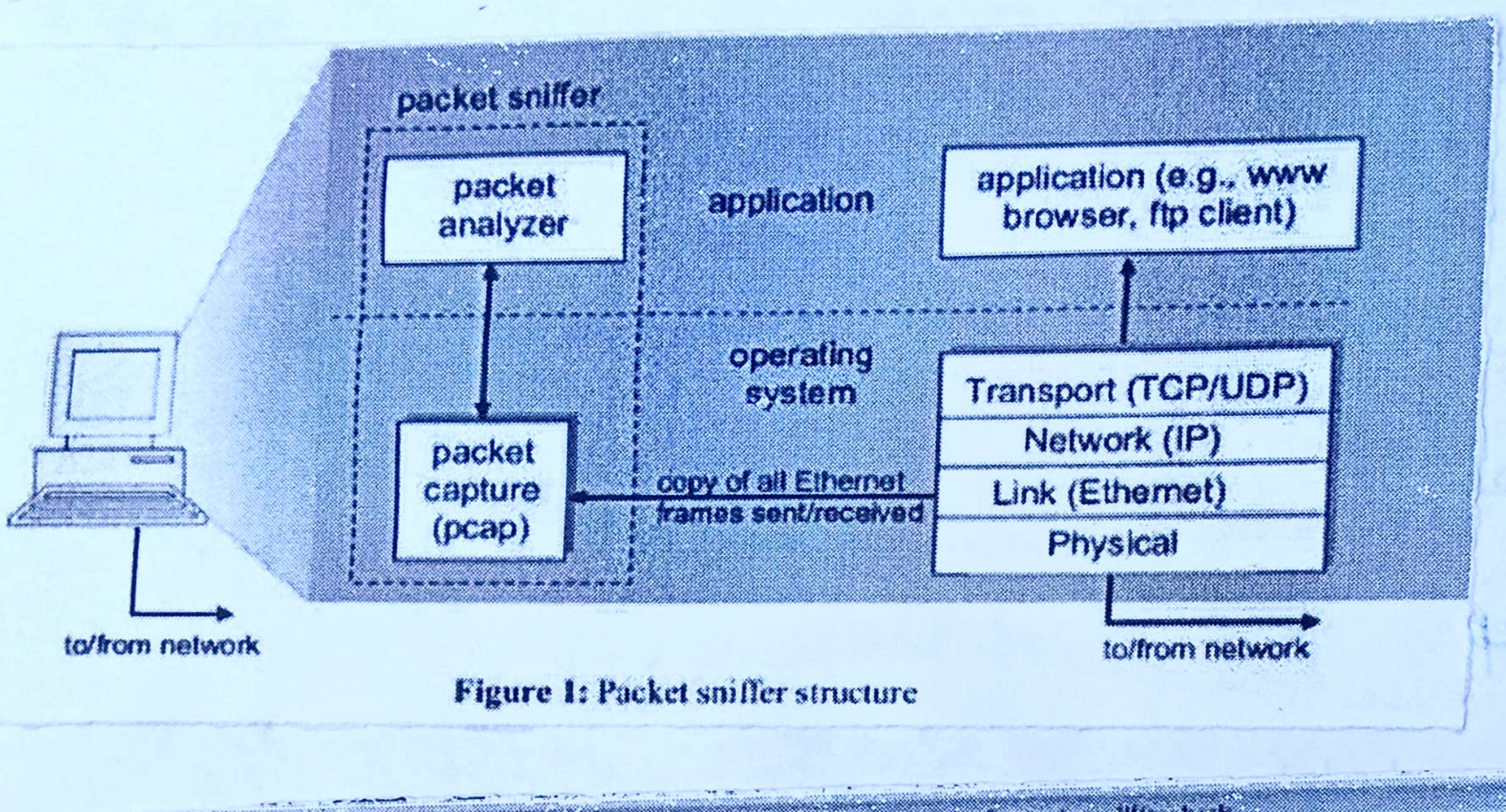
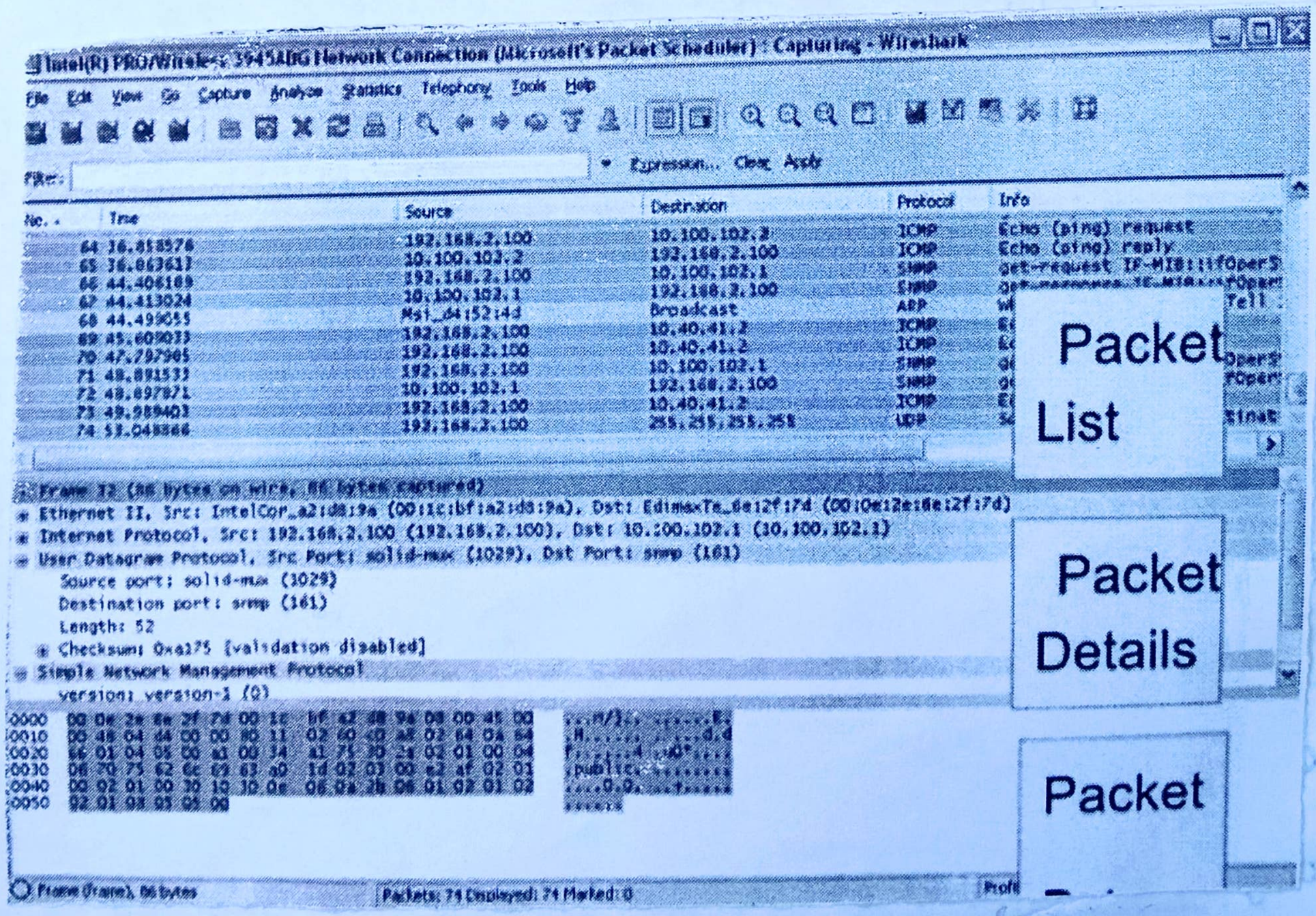


Figure 1: Packet sniffer structure



create a filter to display only DNS packets and provide the flow graph.

→ Go to capture option.

→ Select stop capture automatically after 100 packets.

→ Then click start capture.

→ Search DNS packets in Search bar.

→ TO see flow graph click statistics flow graph.

→ Save the packets.

create a filter to display only HTTP packets and inspect the packets procedure.

→ Select local area connection in wireshark.

→ Go to capture option.

→ Select stop capture automatically after 100 packets.

→ Then click start capture.

→ Search ICMP/IP packets in Search bar.

→ Save the packets

create a filter to display only DHCP packets and inspect the packets procedure.

→ Select local area connection in wireshark.

→ Go to capture option.

→ Select stop capture automatically after 100 packets.

→ Then click start capture

→ Search DHCP packets in search bar.

→ Save the packets.

RESULT:-

Thus the experiments are executed successfully and experiments on packet capture tools wireshark.